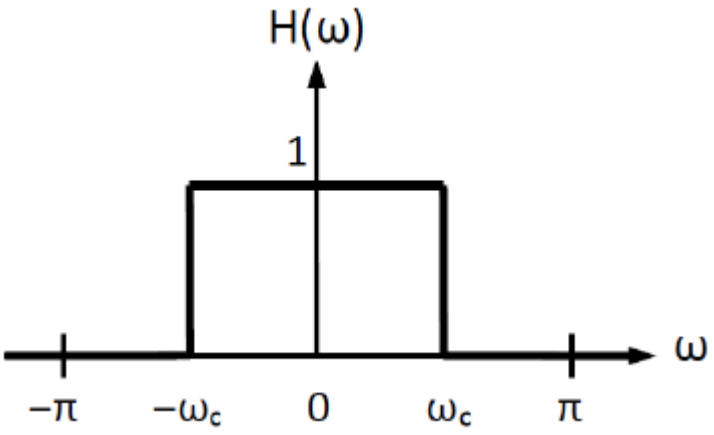
Chebyshev and FIR design

Practice questions · Day 20 · Thu 12 Nov · pack D12

5. Practice

From your course files

T9 Q1 (tutorial). Find the impulse response of the ideal lowpass filter with frequency response shown below. Explain why it is not possible to design an ideal lowpass filter in practice.



T9 Q1 figure (from the tutorial sheet)

T9 Q7 (tutorial). A digital lowpass filter is to designed by applying the bilinear transformation to an analogue Chebyshev type II filter. The digital filter is required to have the following specifications:

Passband ripple: ≤ 1 dB
Stopband attenuation: ≥ 50 dB
Passband edge frequency: 2 kHz
Stopband edge frequency: 4 kHz
Sampling frequency: 40 kHz

Determine ϵ and N for the analogue Chebyshev type II filter.

T9 Q8 (tutorial). Find the missing sample in the following impulse response to make it linear-phase:

$$h(n) = \{\underset{\uparrow}{1}, 2, *, -2, -1\}$$

Show that the resulting FIR system indeed has a linear phase response.

T9 Q9 (tutorial). A causal linear-phase FIR filter of length $M = 5$ is to be designed to approximate the ideal frequency response:

$$H_d(\omega) = \begin{cases} 1, & |\omega| < \pi/6 \\ 0, & \pi/6 \le |\omega| \le \pi. \end{cases}$$

(a) Determine the impulse response of the ideal filter.
(b) Obtain the FIR filter impulse response using the windowing method with a rectangular window.
(c) Repeat part (b) using the Hanning window.

T9 Q4 (tutorial; only if not done in D11). Given the following digital filter specifications, obtain the corresponding analogue filter specifications for bilinear transformation:

$$0.8 \le |H(\omega)| \le 1, \quad 0 \le \omega \le 0.3\pi$$

$$|H(\omega)| \le 0.1, \quad 0.6\pi \le \omega \le \pi.$$

T9 Q2 (tutorial; only if not done in D11). Convert the analogue filter

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using the bilinear transformation.

T9 Q3 (tutorial; only if not done in D11). An ideal analogue integrator is described by $H_a(s) = 1/s$.

(a) Convert the analogue integrator into a digital integrator using the bilinear transformation. Write the difference equation for the digital integrator.
(b) Roughly sketch the magnitude and phase response of the analogue integrator.
(c) Obtain the frequency response of the digital integrator $H(\omega)$, and roughly sketch the magnitude and phase responses.
(d) Compare the magnitude and phase responses in parts (b) and (c), and comment on how well they match.
(e) Considering that the digital integrator has a pole at $z = 1$, what restrictions may need to be placed on the input signal?

New questions

New Q1 [3 marks] An analogue Chebyshev Type I lowpass filter must have at most 0.5 dB passband ripple up to $\Omega_p = 1000$ rad/s, and at least 30 dB attenuation from $\Omega_s = 2000$ rad/s. (a) Find ϵ and N. (b) What order would a Butterworth filter need?

New Q2 [4 marks] Find the poles and the transfer function of a third-order Chebyshev Type I lowpass filter with 1 dB passband ripple and $\Omega_p = 1$ rad/s. At what passband frequencies is $|H| = 1$?

New Q3 [5 marks] A length-4 FIR filter has $h(n) = \{1, 3, *, *\}$ (origin at the first sample). (a) Give the two ways to complete $h(n)$ so that it has linear phase. (b) Which one can be used as a highpass filter? Justify using its zeros. (c) For that filter, write $H(\omega)$ in the form amplitude × linear phase, and state the group delay.

New Q4 [8 marks] A causal linear-phase FIR lowpass filter of length $M = 7$ is to approximate $H_d(\omega) = 1$ for $|\omega| < \pi/2$, 0 otherwise.

a. Find $h_d(n)$ and its values for $|n| \leq 3$. [2]
b. Find $h(n)$ with a rectangular window. [1]
c. Find $h(n)$ with a Hamming window. [2]
d. For both designs, compute $|H(0)|$, $|H(\pi/2)|$ and $|H(\pi)|$. Which window gives the better stopband? [2]
e. Name the linear-phase type and the group delay. [1]

New Q5 [5 marks] The digital lowpass prototype $H_{LP}(z) = \frac{1}{2}(1 + z^{-1})$ has its −3 dB edge at $\omega_p = \pi/2$. Use the L10 digital transformation table to make a highpass filter with passband edge $\omega_p' = \pi/3$. Find $H_{HP}(z)$ and check the gain at $\omega = \pi/3$ and $\omega = \pi$.

New Q6 [6 marks] A digital lowpass filter needs at most 0.5 dB ripple for $0 \leq \omega \leq 0.2\pi$ and at least 40 dB attenuation for $0.35\pi \leq \omega \leq \pi$. It will be designed by applying the bilinear transformation ($T = 1$) to an analogue prototype.

a. Find the pre-warped edges, and ϵ and N for a Chebyshev Type II prototype. [3]
b. How do the Butterworth and elliptic designs compare in order and phase? [1]
c. Using the course Type II response, find the analogue stopband frequencies where $|H| = 0$. [2]

6. Quick recall check

1. Which filter family has the lowest order for a given spec, and which has the best phase?
2. A Chebyshev I filter has $N = 4$ and 2 dB ripple. What is $|H(0)|$ in dB?
3. $h(n) = \{2, *, *, -1, -2\}$. Fill it in to make it linear phase.
4. What is the course Hamming window for $M = 3$?
5. Why can a Type II FIR filter (symmetric, even M) never be a highpass filter?